24CS31 — 3-HOUR HAIL MARY

Organised **unit by unit**, with a map of exactly how each question will be printed and which parts you will be able to answer. Study in the **clock order** below; walk into the hall with the **unit maps**.

THE ONE THING TO UNDERSTAND. "Running total" is what you would score on the **worst** of the four past papers if you stopped studying at that row. You cross the pass mark at **2 h 14**. Two hours of this leaves you at **44**. **Steps 1–12 are the whole plan — finish them.**

THE CLOCK — study in this order

stop anywhere; the total is the worst of 4 papers

#	Study this	Unit	Appears as	Marks	Mins	Clock	Running total
1	**Unitary / Hermitian check**	V	Q10b (5+5 pair)	05	08	0:08	**5**
2	**Prove T is linear + find images**	III	**Q5b**	07	12	0:20	**12**
3	**Least-squares solution**	IV	**Q8b**	10	20	0:40	**22**
4	**t^n-multiplication proof**	I	Q1c / Q2b	07	15	0:55	**29**
5	**Heaviside piecewise + LT**	II	Q3c	07	20	1:15	**36**
6	**$L\{\sin\sqrt{t}\} \to L\{\cos\sqrt{t}/\sqrt{t}\}$**	I	Q1b	04	12	1:27	**40**
7	**All four 2-mark definitions**	I & II	Q1a Q2a Q3a Q4a	02×2	10	1:37	**44**
8	**Orthogonal diagonalization**	V	**Q10c**	10	25	2:02	**49**
9	**$L^{-1}\{\log(\ldots)\}$**	**II**	**Q3b**	**04**	**12**	**2:14**	**52 ✅ PASS**
10	**Span / basis / linear independence**	III	Q5a	06	12	2:26	**53**
11	**Composition of transformations**	III	Q6a	06	12	2:38	**58**
12	**Kernel & range + rank-nullity**	III	Q6c	07	15	2:53	**59**
—— 3 HOURS UP — overtime below, best-paying first ——							
13	Transition matrix / change of basis	III	Q6b	07	15	—	**65** (+6)
14	Positive-definite check	V	Q10a	05	08	—	65
15	Four fundamental subspaces	IV	Q7c	07	15	—	65
16	Complete solution of $Ax = b$	IV	Q7a	06	15	—	65
17	Short inverse LT (partial fractions)	II	Q4b	04	10	—	65

CUT — do not open these. All real, several 4/4, but long and the numbers change every paper: ODEs and simultaneous ODEs • convolution • improper integrals $\int_0^\infty$ • $L\{t\,e^{at}$ trig$\}$ / division by t • periodic functions • $(\sqrt{t}+1/\sqrt{t})^3$ • **QR** • **SVD** • diagonalize + A^n • Gram–Schmidt • projection • Markov • quadratic forms • rotation proof • PCA.
The four calculus items kept (steps 4, 5, 6, 9) survived your own rule — each set with **identical wording or identical values** in at least two papers.

IN THE HALL. Read the two questions in a unit, tick the parts you can do, answer the one with more ticks. **Write every part you know and never leave a unit blank** — a 5-mark check you can do beats a 10-mark decomposition you half-remember. **Answer the parts, not the question.**

UNIT I — Laplace Transforms

20 marks · Q1 *or* Q2 · a(02) + b(04) + c(07) + d(07) · you will score **13**

How the two questions will be printed — layout taken from March 2024, the only paper under your current course title. Green = you have it after the clock above.

	a — definition	b — short result	c — proof / evaluation	d — long evaluation
Q1	**(02)** ✓ Write the LT of a periodic function	**(04)** ✓ Given $L\{\sin\sqrt{t}\}$, show the $\cos\sqrt{t}$ result **step 6**	**(07)** ✓ **t^n-multiplication proof** **step 4**	(07) ✗ $\int_0^\infty$ integral + $L\{t\sin 3t \cos 2t\}$ — *cut*
Q2	**(02)** ✓ Define the Laplace transform	(04) ✗ $L\{(\sqrt{t}+1/\sqrt{t})^3\}$ — *cut*	(07) ✗ $L\{t\,e^{-4t}\sin 3t\}$ and division by t — *cut*	(07) ✗ Periodic function, half-rectifier — *cut*

→ ANSWER Q1 — you have 13 of its 20. The trigger is the t^n proof: in all four papers it sits in the **same question** as the $\sin\sqrt{t}$ result, and the periodic-function problem sits in the **other** one. Find the proof, answer that question, and take the 2-mark definition on your way past.

4 Prove the t^n-multiplication rule

07 marks · Q1c or Q2b — **4/4 papers, word for word**

If $L\{f(t)\} = F(s)$, prove that $L\{t^n f(t)\} = (-1)^n \dfrac{d^n}{ds^n}\{F(s)\}$, n a positive integer.

Appears alongside: the $\sin\sqrt{t}$ result (step 6) — **always, in all four papers**. That pairing is 11 marks in one question.

Script — four lines. $F(s) = \int_0^\infty e^{-st} f(t)\, dt$. Differentiate under the integral sign: $F'(s) = \int_0^\infty (-t) e^{-st} f(t) dt = -L\{t\, f(t)\}$. Each further differentiation drops another factor of $(-t)$. Close with **induction on n** and state the base case $n = 1$.

6 Given $L\{\sin\sqrt{t}\}$, show the $\cos\sqrt{t}$ result

04 marks · Q1b — same statement in Jun 23 and Mar 24

Given $L\{\sin\sqrt{t}\} = \dfrac{\sqrt{\pi}}{2s^{3/2}}\, e^{-1/4s}$, show that $L\{\dfrac{\cos\sqrt{t}}{\sqrt{t}}\} = \dfrac{\sqrt{\pi}}{s^{1/2}}\, e^{-1/4s}$.

Appears alongside: the t^n proof (step 4) — they are never separated.

Method: $\dfrac{d}{dt}(\sin\sqrt{t}) = \dfrac{\cos\sqrt{t}}{2\sqrt{t}}$, so the target equals **2 × L\{that derivative\}**. Apply $L\{f'\} = sF(s) - f(0)$ with $f(0) = \sin 0 = 0$ → $2 \cdot s \cdot \dfrac{\sqrt{\pi}}{2s^{3/2}}\, e^{-1/4s} = \dfrac{\sqrt{\pi}}{s^{1/2}}\, e^{-1/4s}$. **Three lines.**

7 The Unit I 2-mark definition

02 marks · **Q1a and Q2a — one in each, so you get it either way**

“Write the Laplace transform of a periodic function” • “Define the Laplace transform of a function” • “Define a periodic function with an example”

Appears alongside: whichever question you choose — both open with one of these.

Periodic, period T: $L\{f\} = \dfrac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t)\, dt$. **Definition:** $L\{f\} = \int_0^\infty e^{-st} f(t)\, dt$, where the integral converges. **Write the formula, not a paragraph.**

UNIT II — Application of Laplace Transforms

20 marks · Q3 *or* Q4 · a(02) + b(04) + c(07) + d(07) · you will score **13**

	a — definition	b — short inverse	c — main problem	d — application
Q3	**(02)** ✓ Define the Dirac-delta function + sketch	**(04)** ✓ $L^{-1}\{\log(\ldots)\}$ **step 9**	**(07)** ✓ **Heaviside piecewise + LT** **step 5**	(07) ✗ LR-circuit current — *cut*
Q4	**(02)** ✓ Define the unit step function + graph	(04) ✗ Partial-fraction inverse — *overtime 17*	(07) ✗ Verify the convolution theorem — *cut*	(07) ✗ Simultaneous ODEs by LT — *cut*

→ ANSWER Q3 — you have 13 of its 20. The trigger is the **Heaviside piecewise function**. In all four papers an ODE-type problem sits next to it (that is the d part you are skipping) and the convolution question sits in the **other** question. If the log-inverse turns up in Q4 instead, go there — count ticks, don't assume.

5 Heaviside piecewise + Laplace transform

07 marks · Q3c — same function in Apr 23 and Mar 24

Express $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$ in terms of the Heaviside function, and hence find its Laplace transform.

Appears alongside: an ODE-type problem in the *d* slot — in all four papers. You are skipping that 7, deliberately.

Step 1 — stack the jumps: $f(t) = t^2 + (4t-t^2)u(t-2) + (8-4t)u(t-4)$. **Step 2 — rewrite each bracket in ($t-a$):** at $t=\tau+2$, $4t-t^2 = 4-\tau^2$; at $t=\tau+4$, $8-4t = -8-4\tau$. **Step 3 — second shifting** ($L\{g(t-a)u(t-a)\} = e^{-as}G(s)$):
$\boldsymbol{L\{f\} = 2/s^3 + e^{-2s}(4/s - 2/s^3) + e^{-4s}(-8/s - 4/s^2)}$

9 Inverse Laplace transform of a logarithm

04 marks · Q3b — same expression in Apr 23 and Mar 24

Evaluate $L^{-1}\{ \log \dfrac{s^2+1}{s(s+1)} \}$

Appears alongside: the Heaviside question in Mar 24 (both in Q3) — so one question can hand you 11 of these 13 marks.

Method: split → $F(s) = \ln(s^2+1) - \ln s - \ln(s+1)$. Differentiate: $F'(s) = \dfrac{2s}{s^2+1} - \dfrac{1}{s} - \dfrac{1}{s+1}$. Invert term by term → $2\cos t - 1 - e^{-t}$. Then use $L^{-1}\{F'\} = -t\, f(t)$:
answer = $\boldsymbol{\dfrac{1 + e^{-t} - 2\cos t}{t}}$ — the row that takes you past 50.

7 The Unit II 2-mark definition

02 marks · **Q3a and Q4a — one in each**

“Define the Dirac-delta function and sketch its graph” • “Define the unit step function and represent it graphically”

Appears alongside: whichever question you choose — both open with one.

Unit step: $u(t-a) = 0$ for $t < a$, 1 for $t \geq a$ — draw the step at $t=a$. **Dirac delta:** $\delta(t-a)$ is the limit of a unit-area pulse; zero everywhere except $t=a$, with $\int_{-\infty}^{\infty}\delta(t-a)dt = 1$ — draw the spike arrow at $t=a$. **Both are 2 marks for one sentence and one sketch.**

UNIT III — Vector Space and Linear Transformation

20 marks · Q5 *or* Q6 · a(06) + b(07) + c(07) · you will score **13**, or **20** with overtime

This is your strongest unit — zero calculus, and every question type it has ever asked is on this sheet. After overtime step 13 both questions become fully answerable.

	a — vector space (06)	b — transformation (07)	c — the long one (07)
Q5	**(06)** ✓ Span / basis / independence **step 10**	**(07)** ✓ **Prove T is linear + images step 2**	(07) ✗ Rotation-operator proof — *cut*
Q6	**(06)** ✓ **Composition of transformations step 11**	(07) ✗ Transition matrix **overtime 13**	**(07)** ✓ **Kernel & range + rank-nullity step 12**

→ EITHER gives you 13. Q5 = 6 + 7, Q6 = 6 + 7. If you do overtime step 13 (transition matrix, 15 minutes), **Q6 becomes a full 20** — the single biggest jump left in the whole plan, worth +6 on the worst paper. In Apr 23 and Jun 23 the transition matrix was in Q5 instead, so it pays off either way.

2 Prove T is linear + find images

07 marks · **Q5b — all four papers, same slot**

(a) $T(x,y) = (3x+y,\ 2y,\ x-y)$; find the images of (1, 2) and (2, −5).

(b) $T{:}P_2 \to P_2$, $T(ax^2+bx+c) = (a+b)x^2+c$; image of $5x^2+6x+1$.

Appears alongside: the span/basis question at Q5a (step 10) — together they are 13 of Q5's 20.

Method (identical both times): show $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u})+T(\mathbf{v})$ and $T(k\mathbf{u}) = kT(\mathbf{u})$, then substitute the two vectors. **Answers:** $T(1,2) =$ **(5, 4, −1)**, $T(2,-5) =$ **(1, −10, 7)**, $T(5x^2+6x+1) = \mathbf{11x^2 + 1}$.

10 Span / basis / linear independence

06 marks · Q5a — all four papers

Is $\left\{\begin{bmatrix}1&2\\0&1\end{bmatrix}, \begin{bmatrix}3&4\\1&1\end{bmatrix}, \begin{bmatrix}1&2\\1&1\end{bmatrix}, \begin{bmatrix}0&2\\1&2\end{bmatrix}\right\}$ a basis of M_{22}? — or — do {(1,2,3), (−1,−1,0), (2,5,4)} span R^3? — or — is (1,−2) a linear combination of (2,4) and (3,6)?

Appears alongside: the prove-T-is-linear question at Q5b (step 2).

One method covers every version. Flatten each object to a coordinate vector (a 2×2 matrix becomes a 4-vector; a polynomial becomes its coefficient vector), stack them as rows of a square matrix, take the **determinant**. Non-zero ⇒ independent ⇒ basis / spans. Zero ⇒ dependent. For "is **v** a combination of …", solve the system and show it is consistent.

11 Composition of matrix transformations

06 marks · Q6a — all four papers

Determine the matrix for: reflection in the x-axis, then rotation through π/2, then contraction of factor 1/3; find the image of $\begin{bmatrix}4\\1\end{bmatrix}$. (the axis, angle, factor and point change every paper — the method never does)

Appears alongside: the kernel & range question at Q6c (step 12), and the transition matrix at Q6b (overtime 13).

Multiply RIGHT to LEFT: $M = k \cdot R_\theta \cdot Ref$. Reflect in x: $\begin{bmatrix}1&0\\0&-1\end{bmatrix}$ reflect in y: $\begin{bmatrix}-1&0\\0&1\end{bmatrix}$ rotate: $\begin{bmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{bmatrix}$ dilation/contraction by k: scale the whole matrix. For π/2 the rotation is $\begin{bmatrix}0&-1\\1&0\end{bmatrix}$. Then apply M to the point.

12 Kernel and range + verify rank-nullity

07 marks · Q6c — all four papers

Determine the kernel and range of the transformation defined by $A = \begin{bmatrix}1&2&3\\0&-1&1\\1&1&4\end{bmatrix}$, and hence verify the rank–nullity theorem.

Appears alongside: the composition question at Q6a (step 11).

Method: row-reduce A. The **pivot columns of the original A** give a basis for the range; solving $Ax = \mathbf{0}$ gives a basis for the kernel. Here **rank = 2, nullity = 1**, and 2 + 1 = 3 = number of columns. **Also be ready for the formula version** (e.g. $T(x,y) = (3x,\ x-y,\ y)$) — write its matrix first, then the work is identical.

13 Transition matrix / change of basis

07 marks · Q6b — same bases in Sep 23 and Mar 24

$B = \{(1,2), (3,-1)\}$ and $B' = \{(3,1), (5,2)\}$ of R^2. Find the transition matrix from B to B'; if $[u]_B = \begin{bmatrix}2\\1\end{bmatrix}$, find $[u]_{B'}$.

Appears alongside: the composition and kernel&range questions — doing this completes Q6 as a full 20.

Method: put the B' vectors as columns of Q and the B vectors as columns of P. The transition matrix is $\boldsymbol{Q^{-1}P}$, and $[u]_{B'} = (Q^{-1}P)[u]_B$. One 2×2 inverse. **+6 on the worst paper — the biggest single gain left.**

UNIT IV — Orthogonal Projections

20 marks · Q7 (06+07+07) *or* Q8 (10+10) · you will score **10**, or **13** with overtime

Your weakest unit, and that is deliberate — QR and Gram–Schmidt are long and their matrices change every paper, so three hours cannot buy them. You take the least-squares 10 and move on.

	a	b	c
Q7	**(06)** ✗ Complete solution of $Ax\!=\!b$ **overtime 16** (06)	**(07)** ✗ Orthogonal projection of y onto u — *cut* (07)	**(07)** ✗ Four fundamental subspaces **overtime 15** (07)
Q8	**(10)** ✗ QR factorization — *cut* (10)	**(10)** ✓ **Least-squares solution step 3** (10)	**(—)** ✗ —

→ ANSWER Q8 — you have 10 of its 20, from one part. If you reach overtime steps 15 and 16, **Q7 gives you 13** instead and becomes the better choice. Note the four-subspaces question is in **Q7 in all four papers** without exception, so that is where those marks live.

3 Least-squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$

10 marks · **Q8b** — least-squares appears in all four papers

Find the least-squares solution of $AX = b$ for $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$ — identical in Jun 23 and Mar 24

Appears alongside: the QR factorization at Q8a, which you are skipping. Ten marks from one part is still the best available here.

Method: solve the normal equations $A^{\mathrm{T}}Ax = A^{\mathrm{T}}b$. That is the whole question.

Work it once now: $A^{\mathrm{T}}A = \begin{bmatrix} 4 & 8 & 10 \\ 8 & 20 & 26 \\ 10 & 26 & 38 \end{bmatrix}$, $A^{\mathrm{T}}b = \begin{bmatrix} 12 \\ 12 \\ 20 \end{bmatrix}$ $\Rightarrow \boldsymbol{\hat{x} = (10, -6, 2)}$. Substitute back to check — it costs thirty seconds and confirms the whole 10 marks.

15 Four fundamental subspaces

07 marks · **Q7 — in all four papers. Same matrix in 3 of 4.**

Find the dimension and basis for the four fundamental subspaces of $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$ (Jun 23 phrased it "basis for col(A) and nul(A)" — same matrix, same work)

Appears alongside: the complete solution at Q7a (overtime 16) — together they turn Q7 into 13.

Row 3 = row 1, so rank = 2 straight away. dim col(A) = dim row(A) = **2**; dim nul(A) = 4 − 2 = **2**; dim nul(A^{T}) = 3 − 2 = **1**. Row-reduce once and read a basis off each. One of the cheapest 7 marks on the paper.

16 Complete solution of $\boldsymbol{Ax} = \boldsymbol{b}$

06 marks · Q7a — 3 of 4 papers

Find the complete solution of $x + 3y + 3z = 1,\ 2x + 6y + 9z = 5,\ -x - 3y + 3z = 5$.

Appears alongside: the four-subspaces question at Q7c (overtime 15).

Method: row-reduce the augmented matrix, take any particular solution $\boldsymbol{x_p}$, then add the null-space basis. Always present it as $\boldsymbol{x = x_p + c_1 n_1 + \ldots}$ and state the consistency condition. Same row-reduction skill as step 15.

UNIT V — Applications of Eigenvalue Decomposition

20 marks · Q9 *or* Q10 · a(05) + b(05) + c(10) · you will score **15**, or **20** with overtime

The best-value unit on the paper. The two 5-mark property checks are about two minutes of work each, and the 10-marker has used the same matrix in 3 of the 4 papers.

	a — 5-mark check	b — 5-mark check	c — the 10-marker
Q9	(05) Quadratic form, no cross terms — *cut* ✗	(05) Markov steady state — *cut* ✗	(10) SVD — *cut* ✗
Q10	(05) Positive-definite check **overtime 14** ✗	**(05) Unitary / Hermitian check step 1** ✓	**(10) Orthogonal diagonalization step 8** ✓

→ ANSWER Q10 — you have 15 of its 20, and **a full 20** after overtime step 14. The trigger is "orthogonally diagonalize": in 3 of 4 papers the property-checks sit in the same question as it, and SVD sits in the **other** one. In Apr 23 the checks were in Q9 with the property pair — if that happens, take whichever question holds two of your three items.

1 Unitary / Hermitian check

05 marks · Q10b — a property check appears in **all four papers**

Is $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ unitary? — or the Hermitian version — is $\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ Hermitian?

Appears alongside: the positive-definite check — in all four papers the property checks travel as a pair, either as (i)/(ii) of one 10-marker or as the 5+5.

Method: write A^* = the conjugate of the transpose. **Hermitian** ⇔ $A^* = A$. **Unitary** ⇔ $A^*A = I$. **Both of these answer YES** — the 2×2 is unitary (and Hermitian as well); the 3×3 is Hermitian, with a real diagonal and every $a_{ij} = \mathrm{conj}(a_{ji})$. Show A^* written out in full.

8 Orthogonal diagonalization

10 marks · **Q10c — same matrix in 3 of the 4 papers**

Orthogonally diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

Appears alongside: the two 5-mark property checks — in 3 of 4 papers they are in the same question. That is the full 20.

You already know the answer — rehearse producing it. Eigenvalues **2, 3, 6**, all distinct, so the eigenvectors come out mutually orthogonal on their own — <u>no Gram–Schmidt needed</u>, and saying so earns marks.

Eigenvectors $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ for 2, $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ for 3, $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ for 6. Normalise by √2, √3, √6 — those are the columns of P. Then $\boldsymbol{P^TAP} = \mathbf{diag(2, 3, 6)}$. Verify all three pairwise dot products are 0 in your answer.

14 Positive-definite check

05 marks · Q10a — same matrix in Apr 23 and Mar 24

Check whether $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$ is positive definite or not.

Appears alongside: the unitary/Hermitian check (step 1) — doing both completes Q10's 5+5 and makes the unit a full 20.

Method: leading principal minors. They are **1, 0, 0**, so it is **NOT positive definite** — it is positive *semi*-definite (row 3 = row 1, eigenvalues 0, 0, 6). **Say that explicitly**; writing only "no" loses marks. Two minutes of work.

17 Short inverse Laplace transform

04 marks · Q4b — a short inverse appears in **both** Unit II questions

Find $L^{-1}\{\frac{3s+2}{s^2-s-2}\}$

Method: factor the denominator $(s-2)(s+1)$, split into partial fractions, and invert with the standard table. This is insurance for the case where the log-inverse (step 9) turns up in the question you did *not* choose — one of the two 4-mark inverses is always reachable.

How this was built: each candidate topic was costed in study minutes and scored by the marks it adds to the *worst* of the four past papers, respecting the fact that only one question per unit can be answered. The list is greedy-ordered by worst-case marks per minute, so stopping at any row leaves the best score reachable in the time spent. Question maps use the March 2024 layout — the only paper printed under your current course title. Running totals are computed, not estimated; the working is in *coverage_audit.py*.